

GEOOptimize

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Proposal for:	Proposal Date:	Proposal Number:
Cormac Foster, P.Eng. Vice President 2175 Mountain Rd Dauphin, MB R7N 3K5	2026-03-18	QU-0224
		Reference
		CA MB Vermillion Growers

Understanding of Project

Vermillion Growers, a participant in the Manitoba Green Advantage Program administered by the Manitoba Chambers of Commerce, is evaluating opportunities to improve the sustainability and energy performance of its operations. As part of this initiative, the company is exploring the potential implementation of ground source heat pump (GSHP) systems as an energy-saving measure for its facility.

The facility is located at 2175 Mountain Road in Dauphin, Manitoba, and consists of an approximately 12-acre greenhouse and a 17,500 square foot warehouse. Vermillion Growers is currently working with Efficiency Manitoba to complete a commercial energy audit to further assess energy use and identify efficiency opportunities across the site.

The warehouse has previously been evaluated for a GSHP system by Epp Siepman Engineering, which developed a conceptual design incorporating horizontal ground loops beneath an existing retention pond. However, this system has not yet been implemented due to capital cost constraints during initial facility development.

Vermillion Growers is now seeking a feasibility study to evaluate the applicability of a GSHP system for the warehouse, the greenhouse, or both. In particular, the customer is interested in understanding whether a geothermal system could be effectively applied to a greenhouse of this scale, either as a primary heating solution or in a supplementary capacity.

Scope of Work

Based on preliminary discussions and a review of available information, it has been determined that evaluating a GSHP system for the 12-acre greenhouse is not warranted as part of this study. The greenhouse operation relies on supplemental CO₂ to support plant growth, which is currently supplied by directing flue gas from natural gas-fired boilers into the growing space.

Transitioning to a GSHP-based heating system would eliminate this source of CO₂, necessitating the installation of an alternative CO₂ supply system. The additional capital and operating costs associated with CO₂ supplementation would significantly reduce, or potentially negate, the economic viability of a geothermal solution for the greenhouse. As such, this study will focus exclusively on the warehouse facility.

The scope of work for the warehouse GSHP feasibility assessment will include the following:

- **Energy Modeling:** Develop a detailed hourly energy model of the warehouse using data and findings from the Efficiency Manitoba commercial energy audit to characterize current and projected heating and cooling loads.
- **Geotechnical and Thermal Assessment:** Review available geological and site data to estimate subsurface thermal properties and evaluate ground heat exchanger options, including identification of cost-effective drilling depths and installation considerations based on local contractor practices.
- **System Design Review:** Review and refine the existing GSHP system concept developed by Epp Siepmann Engineering, including validation of assumptions and alignment with current facility requirements and operating conditions.
- **System Costing:** Develop current capital cost estimates for the proposed GSHP system, including ground loop field and mechanical system components.
- **Economic Analysis:** Perform a life-cycle cost analysis (LCCA) comparing the proposed GSHP system to the existing or alternative heating systems. This analysis will include evaluation of operational costs, return on investment (ROI), net present value (NPV), and simple payback.

Information Required

- Detailed construction drawings and specifications for the proposed building, including:
 - Insulation values of walls, floor, and roof.
 - Window specifications (solar heat gain coefficient (SHGC) and U-value).
 - Estimated occupancy schedule of the building.
 - Proposed mechanical system for the building, including ventilation requirements and specifications.
 - Electrical and lighting drawings and specifications for the building.
- Efficiency Manitoba energy audit.
- Site plan indicating land area available for construction of the GHX.
- Geotechnical report.
- Existing GSHP system construction drawings.

Deliverables

- A comprehensive feasibility analysis, provided in PowerPoint format, summarizing system viability, key assumptions, design considerations, and preliminary findings.
- AutoCAD drawing package for the proposed GHX including:
 - GHX site plan illustrating borefield layout and spatial constraints
 - GHX system details depicting conceptual design elements and configuration
- All drawings are intended for conceptual and planning purposes only and will be issued as non-stamped, non-sealed documents.

Pricing

Description			Line Total
Feasibility Study			\$18,450.00
Total (CAD)			\$18,450.00

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